

TITAN SMALL BUSINESS SERVICES: SECURITY ARCHITECTURE DOCUMENT (SAD)

Operator: [James Emanuel] **Date:** [04/15/2026]

1. Perimeter Hardening (UFW & SSH)

- **SSH Status:** [Commands used to disable root login and password: `sudo nano /etc/ssh/sshd_config` , Settings changed: `PermitRootLogin no` , `PasswordAuthentication no` , `sudo systemctl restart ssh` , Tell Linux this file is allowed to run as a program: `chmod +x -/dc_auditor.py`]
- **Firewall Logic:** [`sudo ufw default deny incoming` , `sudo ufw allow 22` , `sudo ufw allow 8080` , `sudo ufw enable`]

2. The Automated Auditor (Python)

- **Script Logic:** [

```
python import os response = os.system("ping -c 4 [YOUR_DC_IP]") with open("/var/log/dc_audit.log", "a") as log: if response == 0: log.write("DC is UP\n") else: log.write("DC is DOWN\n")
```

]
- **Telemetry Path:** `/var/log/sys_audit.log`

3. Containerized App (Docker)

- **Network Isolation:** [The application is deployed using two isolated networks: “frontend” and “backend”. The database is hidden in the backend where the internet can’t touch it]
- **Stack Health:** [NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS cybersecurity-phase1_web_1 wordpress:latest “docker-entrypoint.s...” web 49 minutes ago Up 41 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp]

4. Executive Summary

[The overall security posture of the outpost is defined by a Defense in Depth strategy, ensuring that multiple layers of protection exist between an attacker and the sensitive data. By implementing an “identity-first” SSH lockdown and a strict “Default Deny” firewall, the perimeter is hardened against unauthorized external access and common brute-force attacks. Finally, the use of container isolation and automated system auditing ensures that even if one service is compromised, the backend database remains air-gapped and internal threats are logged in real-time.]